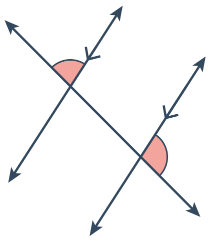
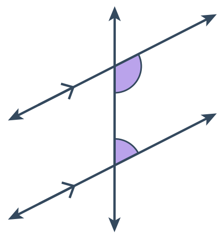
# Parallel lines and transversals

**1** Identify the types of angles formed by the transveral in the following diagrams:

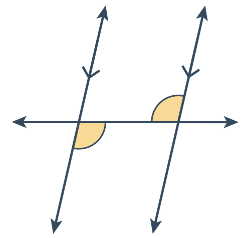
**a**



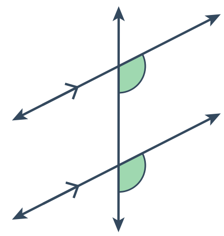
**b**



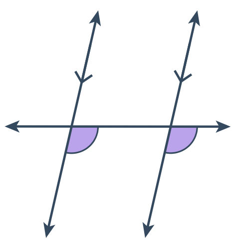
**c**



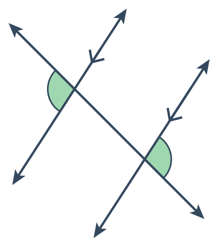
**d**



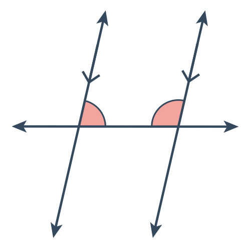
**e**



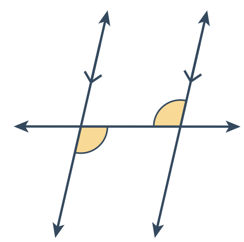
**f**



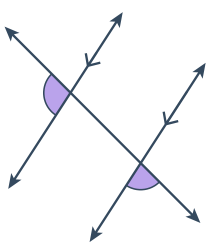
**g**



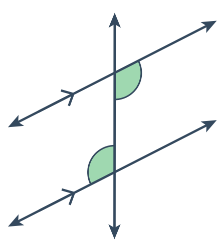
**h**



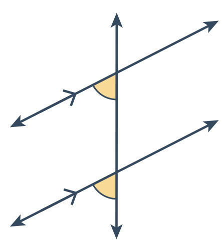
**i**



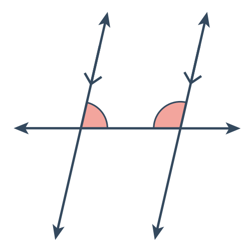
**j**



**k**



**l**

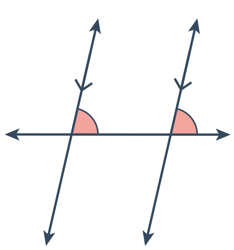


**2** Consider the following diagrams:

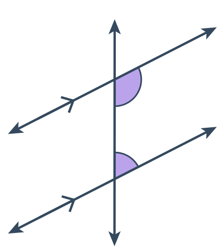
**i** State whether the two highlighted angles are congruent, complementary or supplementary.

**ii** Identify the types of angles formed by the transveral.

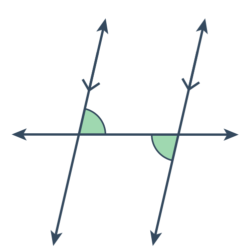
**a**



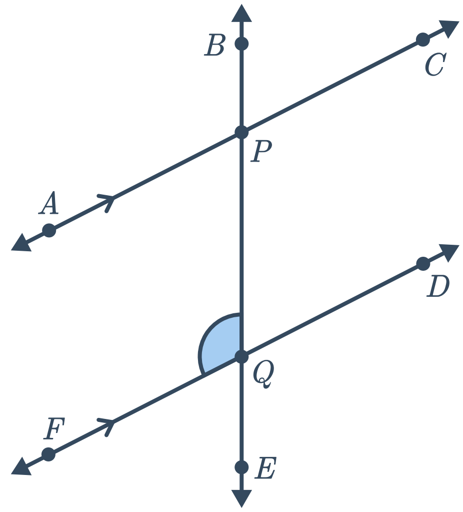
**b**



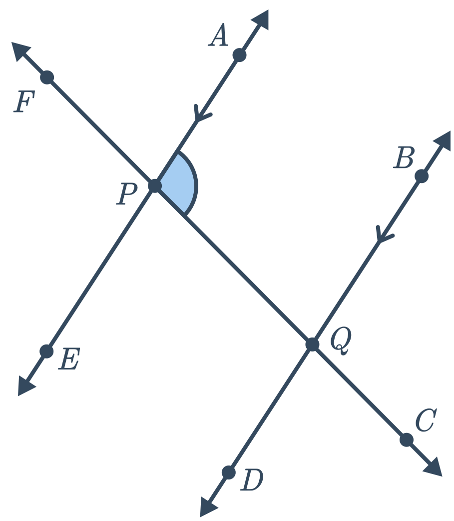
**c**



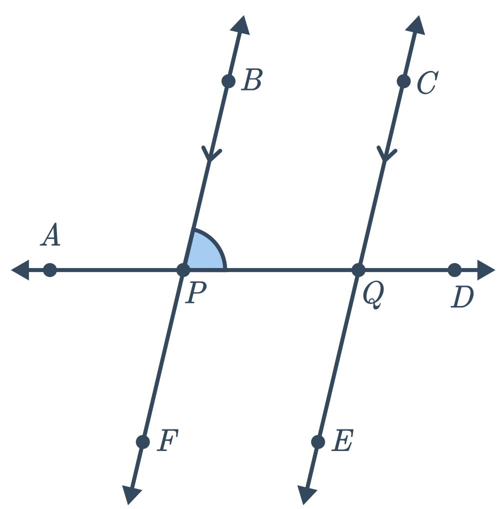
**3** Consider the given diagram:
Write the angle that is corresponding to $\angle BQF$.




**4** Consider the given diagram:
Write the angle that is alternate interior to $\angle APC$.

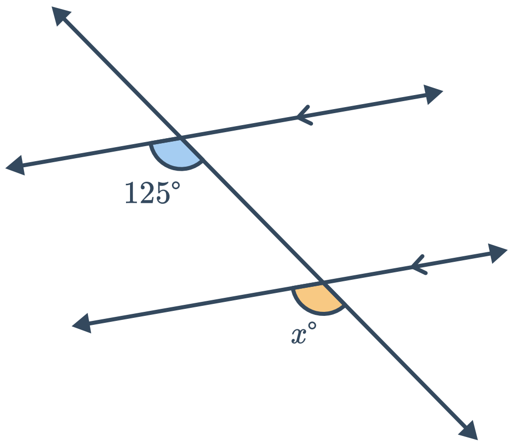



**5** Consider the given diagram:
Write the angle that is consecutive interior to $\angle BPD$.

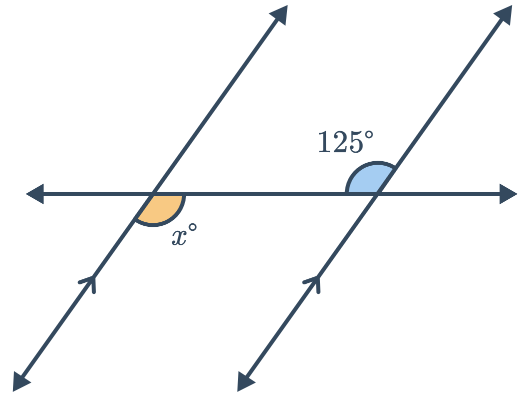


**6** For the following, solve for the value of $x$.

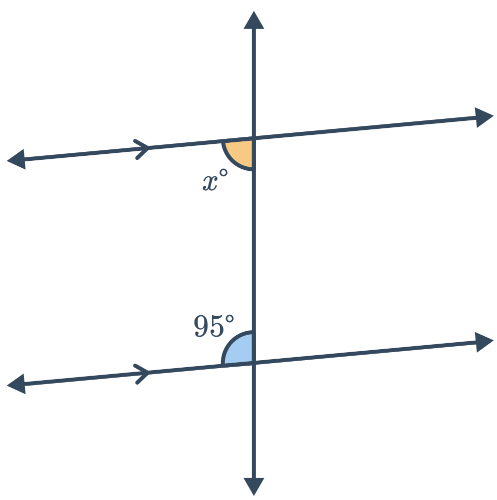
**a**



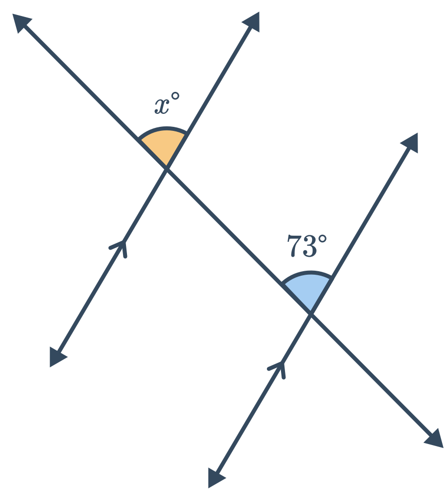

**b**



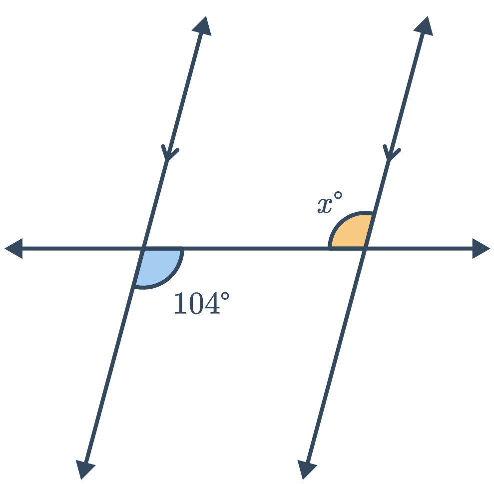

**c**



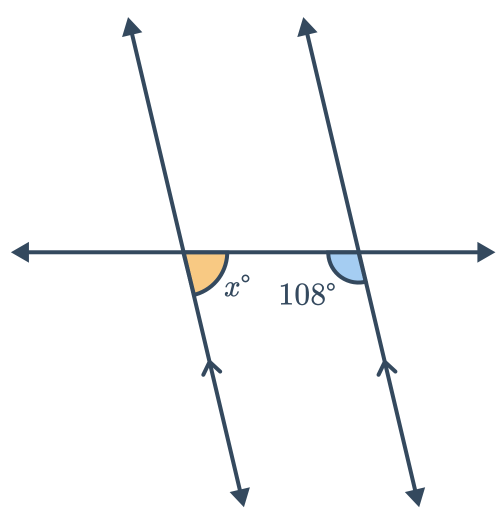

**d**



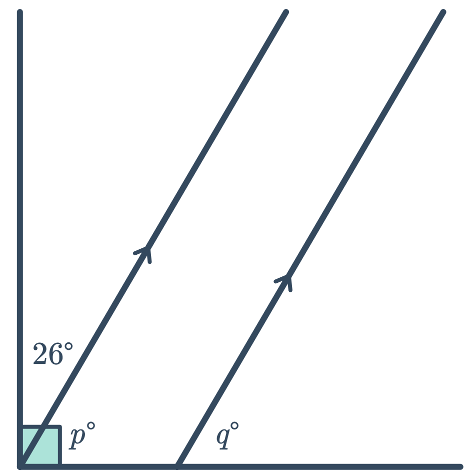

**e**




**f**



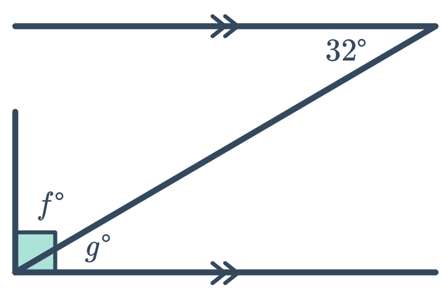
**7** Consider the given diagram:




**a** Solve for $p$.

**b** Solve for $q$.

**8** Consider the given diagram:

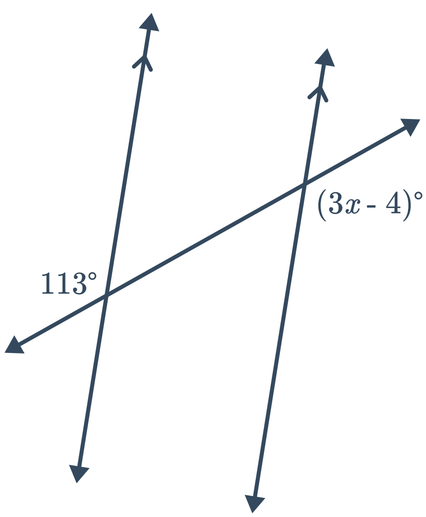



**a** Solve for $g$.

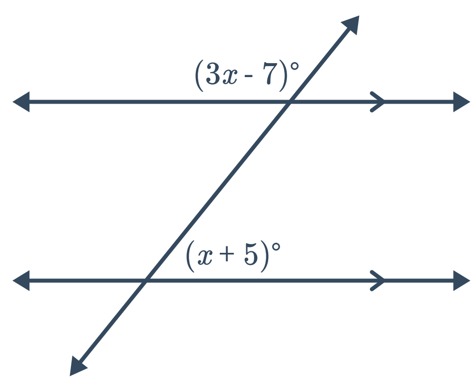
**b** Solve for $f$.

**9** For the following, solve for $x$.

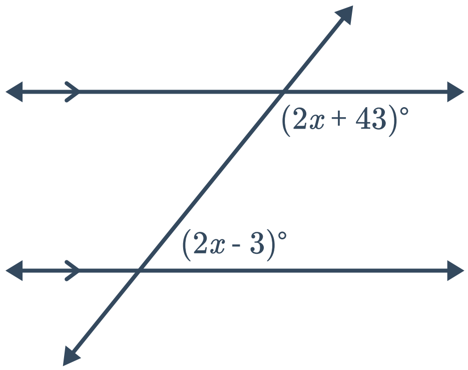
**a**




**b**



c

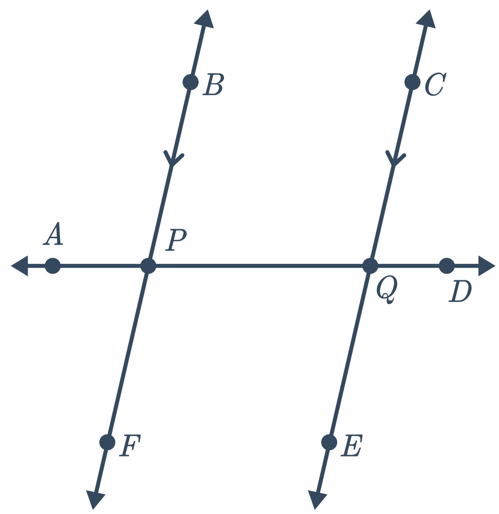



**10** Consider the given diagram:

a How are $\angle DPF$ and $\angle AQC$ related?

b How are $\angle AQC$ and $\angle CQD$ related?

c If $\angle DPF = 102^\circ$, find $m\angle CQD$.

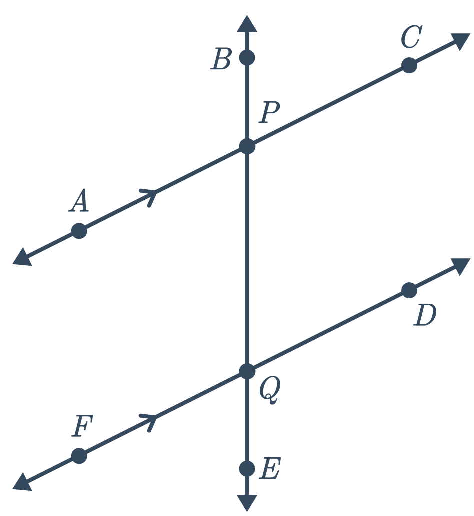



**11** Consider the given diagram:

a How are $\angle APB$ and $\angle CPE$ related?

b How are $\angle CPE$ and $\angle BQD$ related?

c If $m\angle APB = 106^\circ$, find $m\angle BQD$.

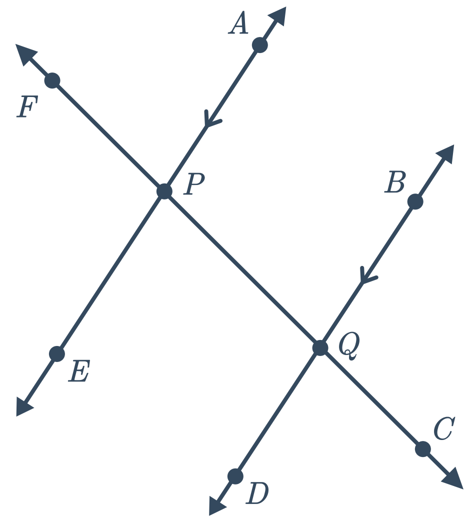


**12** Consider the given diagram:

**a** How are $\angle DQF$ and $\angle CPE$ related?

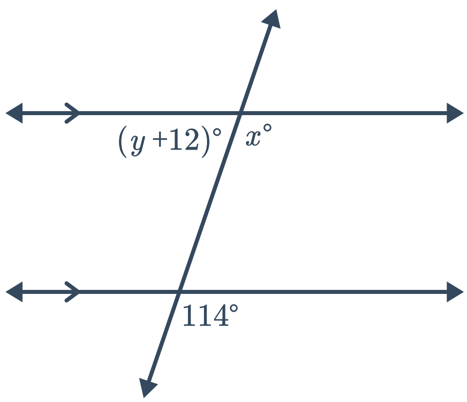
**b** How are $\angle CPE$ and $\angle EPF$ related?

**c** If $m\angle DQF = 102^\circ$, find $m\angle EPF$.

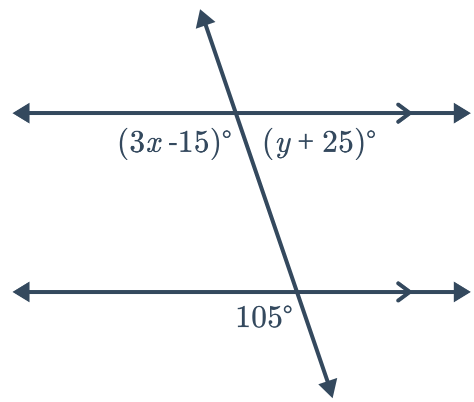



**13** For the following diagrams, solve for the value of $x$ and $y$.

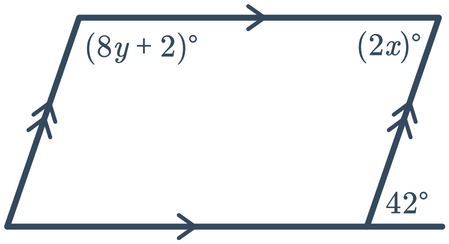
**a**



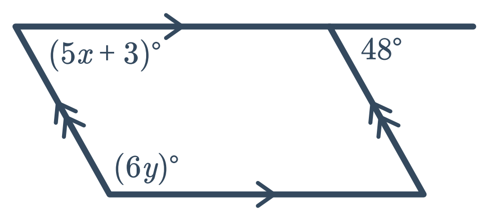

**b**




**c**

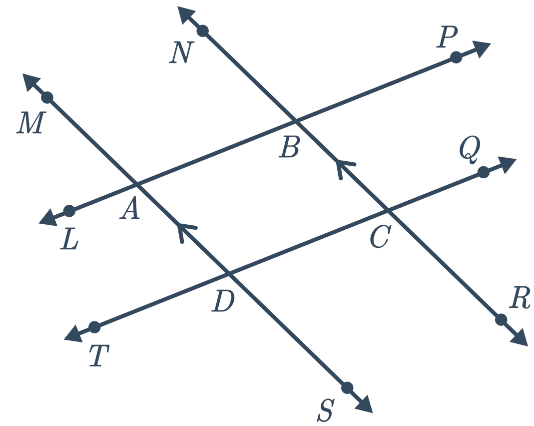



**d**



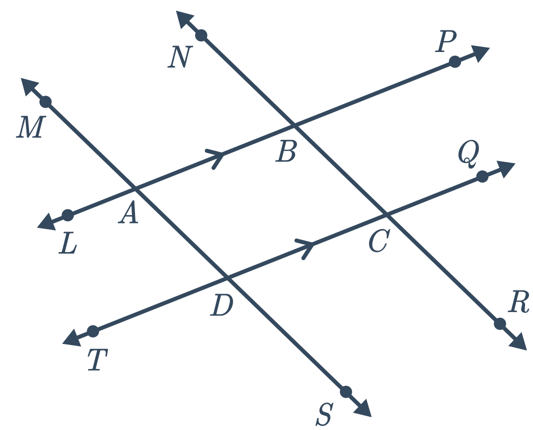
**14** Consider the following diagram:
Given that $\overleftrightarrow{AD} \parallel \overleftrightarrow{BC}$, which of the following are pairs of congruent angles?

- $\angle ADC$ and $\angle DCR$
- $\angle NBA$ and $\angle QCR$
- $\angle BAD$ and $\angle ADT$
- $\angle ABC$ and $\angle BAD$



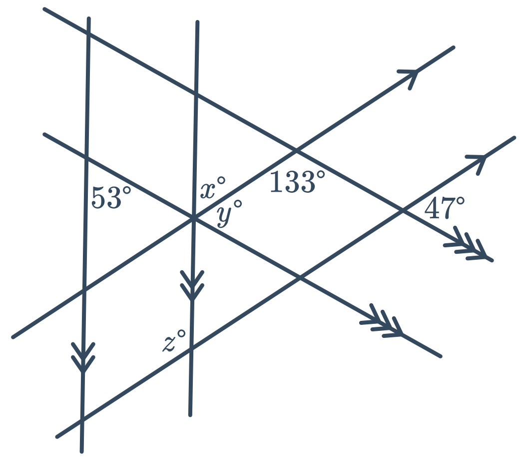

**15** Consider the following diagram:
Given that $\overleftrightarrow{AB} \parallel \overleftrightarrow{DC}$, which of the following are pairs of congruent angles?

- $\angle NBP$ and $\angle DCR$
- $\angle BCQ$ and $\angle TDS$
- $\angle NBA$ and $\angle BAD$
- $\angle MAL$ and $\angle TDS$



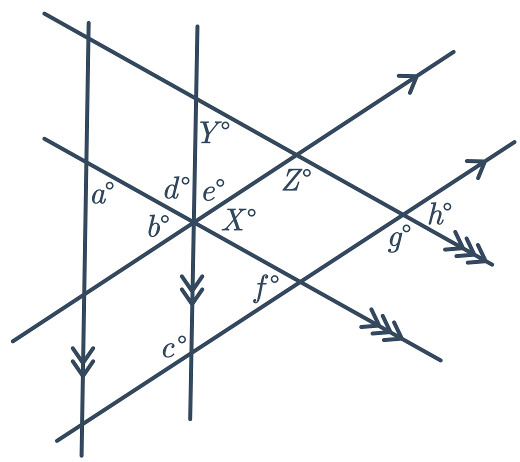
**16** Consider the given diagram, which has thre[redacted]s of parallel lines.
Find the value of the following:

**a** $y$ **b** $x$ **c** $z$




**17** Consider the given diagram, which has three pairs of parallel lines.
Write all variables that have the same value as the following:

**a** $X$ **b** $Y$ **c** $Z$

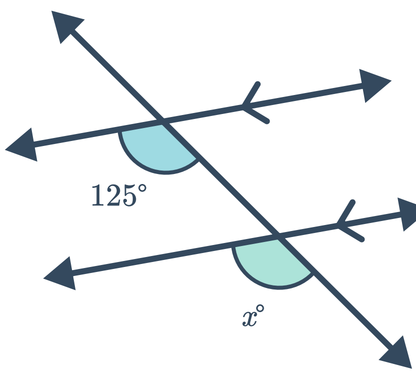


## Additional questions

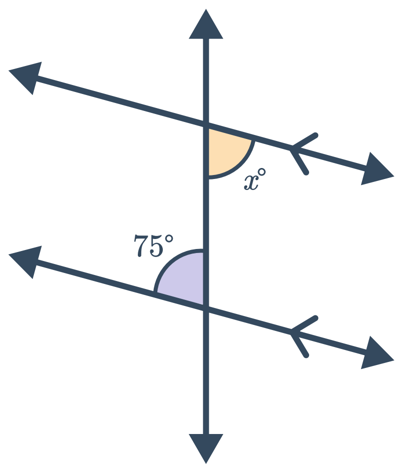
**18** For each of the following diagrams:

- **i** Identify the types of angles formed by the transveral in the following diagrams
- **ii** Describe how the highlighted angles are related to each other
- **iii.** Find the value of $x$

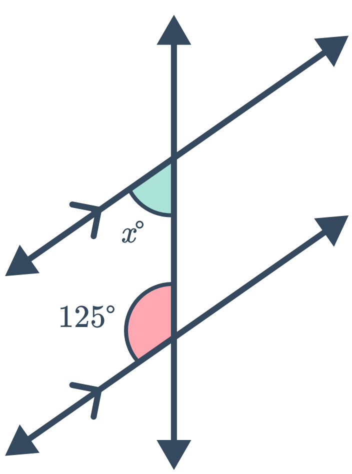
**a**



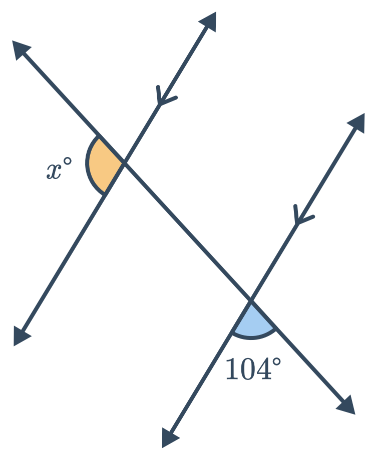

**b**



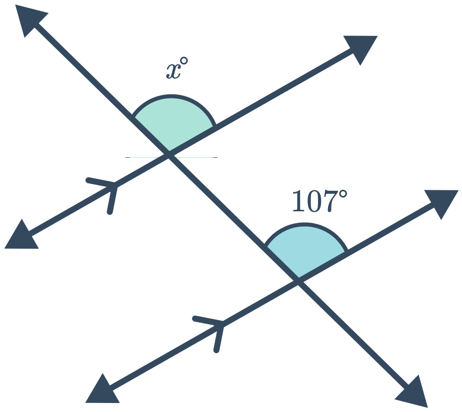

**c**

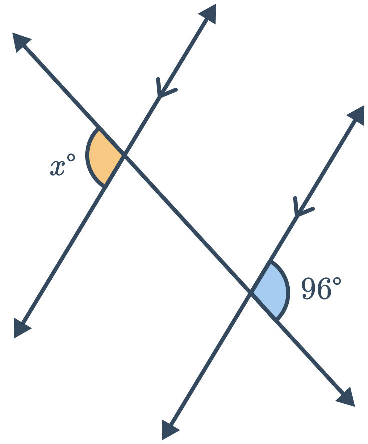



**d**

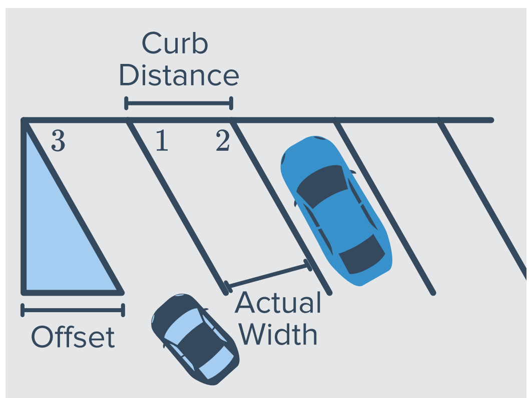


e



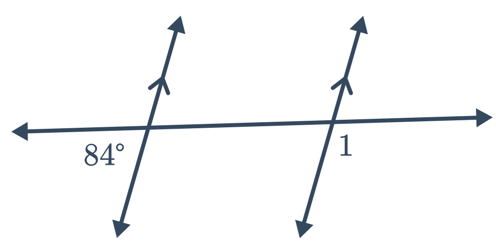




19 In the figure of the parking lot shown, the lines from the parking spaces should be parallel. If $m\angle 3 = 51^\circ$, what should $m\angle 1$ and $m\angle 2$ be? Explain your reasoning.

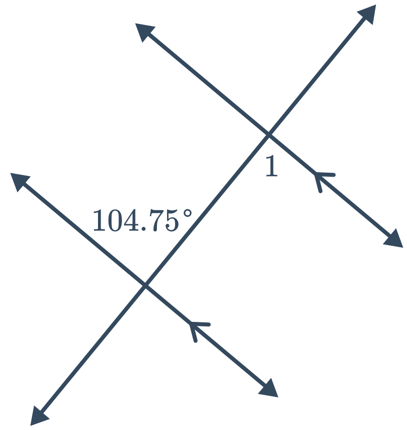



20 Review the following diagrams and statements. Determine whether the statements are correct. Explain your reasoning.

a $m\angle 1 = 84^\circ$ by the consecutive exterior angles theorem

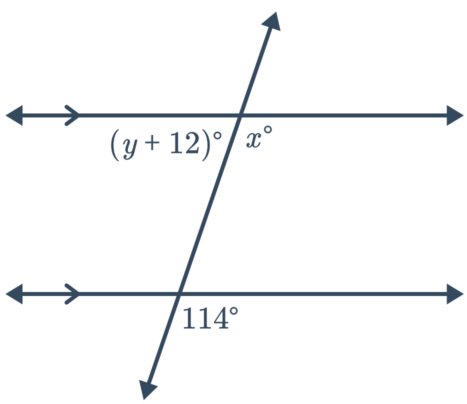


**b** $m\angle 1 = 75.25^\circ$ by the alternate interior angles theorem

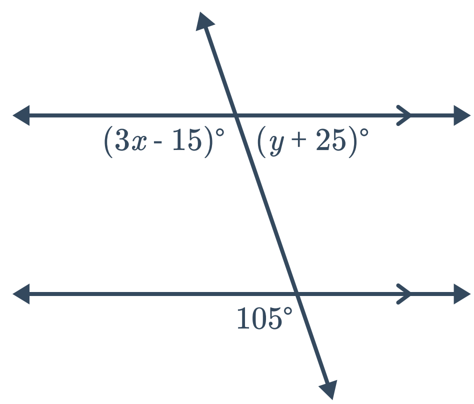



**21** Consider the following diagrams:
Find the value of the unknown variable(s)

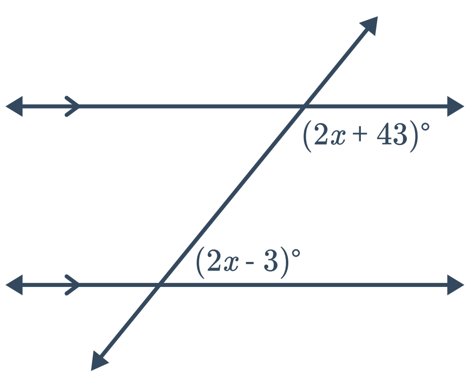
**a**



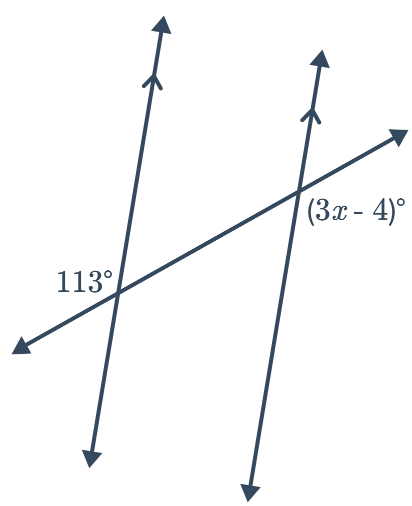

**b**



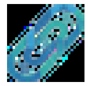

**c**

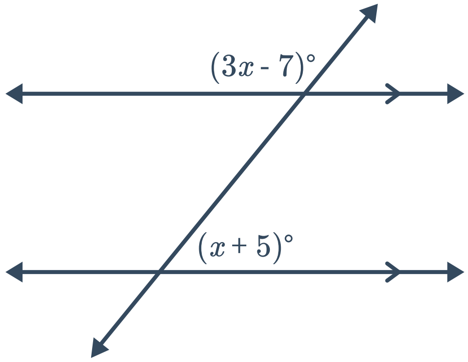


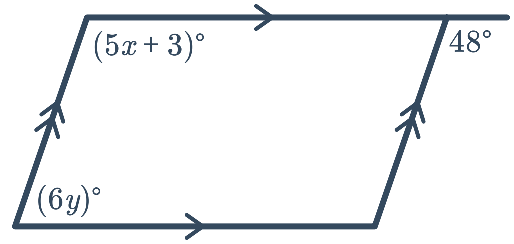

**d**



e



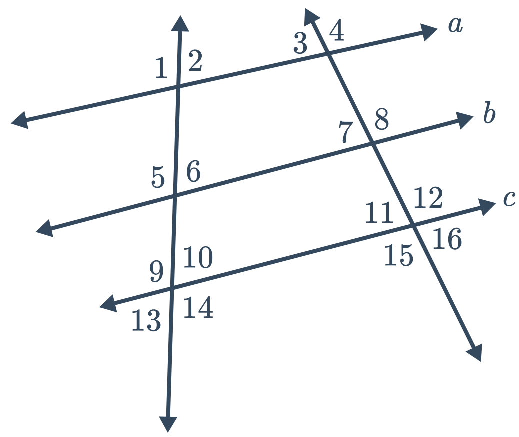


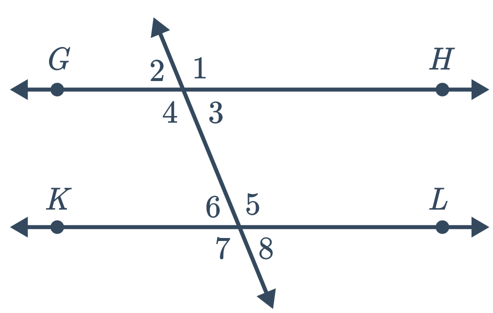

**22** Determine if the information given is enough to justify the conclusion.

**a** Given: $a \parallel b$ and $\angle 1 \cong \angle 9$
Conclusion: $\angle 5 \cong \angle 9$

**b** Given: $\angle MNO$ and $\angle ONP$ are corresponding angles
Conclusion: $\angle MNO \cong \angle ONP$



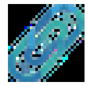

**23** Prove the alternate exterior angles theorem:
Given: $\overleftrightarrow{GH} \parallel \overleftrightarrow{KL}$
Prove: $\angle 2 \cong \angle 8$

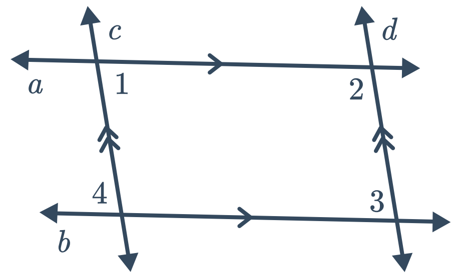


**24**

**a** Given: $a \parallel b, c \parallel d$
Prove: $\angle 1 \cong \angle 3$

**b** Prove part (a) in a different way.






**25** Given: $\overleftrightarrow{AF} \parallel \overleftrightarrow{GL}$ and $\angle BHJ \cong \angle HJE$
Prove: $\angle HBE \cong \angle BEJ$